

Invent a Creature

Mode 2 — build a whole new character family (for teens) · 模式二：创造全新的角色家族（适合青少年）

Clawd is a **rect-crab**. In this guide you'll create a sibling species — say, a **rect-bird** — as a *grammar pack*: a folder that teaches the AI everything about your creature. Once your pack exists, anyone (including the 8-year-olds using the Mode 1 guide) can make unlimited animations of YOUR creature with one command.

Clawd 是“方块蟹”。在这份指南里，你将创建一个新物种——比如“方块鸟”——做成一个**语法包**：一个教会 AI 你的生物一切的文件夹。语法包做好后，任何人（包括用模式一的弟弟妹妹）都能用一条命令给你的生物做无数动画。

What's in a grammar pack 语法包里有什么

File	What it is
<code>grammar.json</code>	Name card: id, character name, file prefix, canvas size 名片：编号、角色名、文件前缀、画布尺寸
<code>base.svg</code>	The puppet: your creature drawn once, standing still, every part named 木偶：你的生物的静止姿势，每个部件都有名字
<code>rules.md</code>	The law book: colors, shapes allowed, how animations must work 法律书：颜色、允许的形状、动画规则
<code>examples/*.svg</code>	At least one finished animation 至少一个完成的动画
<code>PLANS.md</code> + <code>plans-template.md</code>	Optional: idea briefs and a blank Idea Card 可选：创意简介和空白创意卡

Full technical contract: `animation-studio/SPEC.md` , section 2.

1

Design your puppet on paper 在纸上设计木偶

Grid paper, roughly a 15×16 box, **rectangles only**. Your creature needs: a body, at least two limbs that can move, eyes, and a ground shadow. Budget: **6–12 rectangles**. Fewer than 6 and it can't show feelings; more than 12 and the AI starts losing track of parts.

Give it one unforgettable silhouette trait. Crab = wide body + side claws. A bird might be: tall body, a 2×1 beak sticking out, wing rectangles at the sides, stick legs.

用方格纸，在大约 15×16 的格子里，**只用长方形**设计。必须有：身体、至少两个会动的部件、眼睛、地面影子。预算 6–12 个长方形。给它一个一眼就认出的轮廓特征：螃蟹是宽身体加两侧钳子；鸟可以是高身体、凸出的 2×1 嘴巴、两侧翅膀、细腿。

2

Build `base.svg` 制作基本姿势

Copy `grammars/rect-crab/base.svg` into a new folder `grammars/rect-bird/` and reshape it rectangle by rectangle. Rules of thumb:

- One `<rect>` per body part, each with an `id` named after the part AND its joint: `left-wing` (pivots at the shoulder), `beak` (pivots where it meets the head).
- Group same-colored parts: `<g fill="#F2B84B">...</g>`.
- Keep a ground line (crab uses `y=15`) and a shadow rect on it.
- Check your work by double-clicking the file — the whole creature should be visible, nothing cut off.

把 `grammars/rect-crab/base.svg` 复制到新文件夹 `grammars/rect-bird/`，一块一块地改。每个部件一个 `<rect>`，id 用“部件+关节”命名（如 `left-wing` 在肩膀转动）。同色部件放进一个 `<g>`。保留地面线和影子。双击检查：整个生物都要看得见。

3

Write the law book: `rules.md` 写法律书

Copy rect-crab's `rules.md` and edit the parts that describe the character; **keep every generic rule** (CSS keyframes only, seamless loops, rectangles only, file naming, the checklist). You must update:

- Body color(s) and eye spec — exact hex codes and positions.
- The part list with their joints.
- The animation canvas: pad your base viewBox ~15 units left/right and ~25 up for effects (crab: `"-15 -25 45 45"`).
- Creature-specific physics! Crab rules say legs stay planted. A bird's rules should say the opposite: *"the bird may leave the ground; wings flap by rotating at the shoulder."* This one line changes everything the AI makes.

复制方块蟹的 `rules.md`，只改描述角色的部分；**通用规则全部保留**（只用 CSS 关键帧、无缝循环、只用长方形、文件命名、检查表）。必须更新：身体和眼睛的颜色位置、部件及关节清单、画布大小、生物专属的物理规则——螃蟹“腿不离地”，鸟应该写“可以离地，翅膀在肩膀处转动扇动”。这一行会改变 AI 做的一切。

4

Fill in `grammar.json` 填名片

```
{
  "schemaVersion": 1,
  "id": "rect-bird",
  "name": "Rect-Bird (Pip)",
  "characterName": "Pip",
  "filePrefix": "pip",
  "tier": "rect",
  "animationViewBox": "-15 -25 45 45",
  "outputWidth": 500,
  "outputHeight": 500
}
```

`id` must equal the folder name. `filePrefix` is how output files get named: `pip-first-flap.svg`.

`id` 必须和文件夹名一样。 `filePrefix` 决定输出文件名。

5 The real test: make its first animation 真正的考试：第一个动画

Run your own pack through Mode 1:

```
python3 tools/build_prompt.py first-flap "Pip flaps both wings and hops in place, feathers drifting down" --grammar rect-bird
```

Paste the prompt into `claude.ai` (or use `claude_animate.py`). If the result is off-model — wrong colors, redrawn body, no flapping — **the bug is in your `rules.md`**, not in the AI's answer. Add the missing rule and try again. When an animation passes the checklist, save it into `grammars/rect-bird/examples/`. Done — from now on every future prompt includes it as the worked example, and your grammar gets dramatically more reliable.

用模式一测试你自己的语法包。如果结果不对——颜色错了、身体被重画、翅膀不动——**问题出在你的 `rules.md`**，不是 AI。补上缺的规则再试。通过检查表后，把动画存进 `examples/`。以后每次生成都会带上它做示范，你的语法包会可靠得多。

6 Definition of done 完成标准

- ☐ `build_prompt.py --grammar rect-bird` runs with no errors.
- ☐ `base.svg` shows the whole creature in a browser.
- ☐ Every movable part has its own named `rect/group`.
- ☐ `rules.md` covers: canvas, colors, parts+joints, physics, looping, naming, checklist.
- ☐ At least one example animation loops seamlessly in a browser.
- ☐ A one-sentence Mode 1 request produces a recognizable, on-model animation.

六条全部打勾，你的新物种就正式诞生了！

★ Advanced: curve grammars (anime style) 进阶：曲线语法 (动画风)

Rect grammars are the beginner tier. A **curve-tier** grammar (set `"tier": "curve"`) may use paths, circles, and rounded shapes — that's the door to a 1990s-anime-style character. Everything above still applies, plus your `rules.md` must also pin down:

- **Outlines:** stroke color and width on every shape (e.g. `stroke="#222" stroke-width="0.6"`) — outlines ARE the anime look.
- **Cel shading:** each color gets exactly one flat shade tone. No gradients — still banned.
- **Signature marks:** blush marks, eye sparkles — define them as reusable shapes with exact positions.

Ship 2–3 examples instead of 1 — curved styles need more evidence for the AI to copy. Expect more iterations. The animation rules never change: CSS keyframes, grouped parts, seamless loops.

方块语法是入门级。**曲线级**语法 (`"tier": "curve"`) 可以用路径、圆形、圆角——这是通往 90 年代动画风角色的大门。以上全部照做，另外 `rules.md` 还必须规定：描边（颜色和粗细——描边就是动画感的来源）、赛璐璐上色（每种颜色只配一个纯色阴影，仍然禁止渐变）、招牌记号（腮红、眼睛里的高光——写清楚确切位置）。示例要给 2–3 个，迭代次数也会更多。动画规则永远不变：CSS 关键帧、分组部件、无缝循环。